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A photograph of a young man with dark hair, wearing a grey zip-up sweater over a white collared shirt. He is looking upwards towards a large digital screen that fills most of the background. The screen displays several lines of C code, which appears to be a fast Fourier transform (FFT) implementation. The code includes comments and mathematical operations involving matrices and trigonometric functions. The lighting is dim, focusing attention on the man and the bright screen.



FRAMA-C's E-ACSL Plug-in

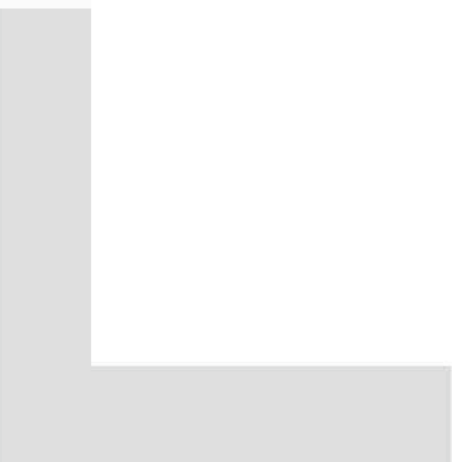
Release 0.1+dev compatible with Fluorine-20130501

Julien Signoles

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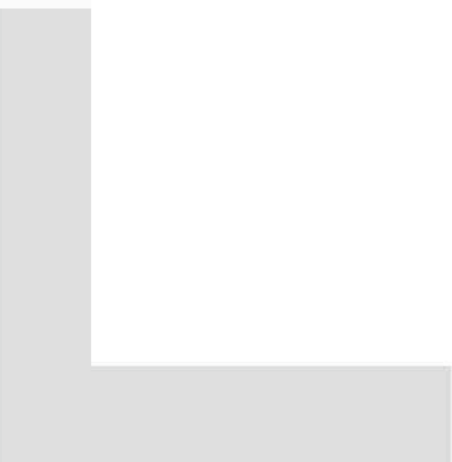
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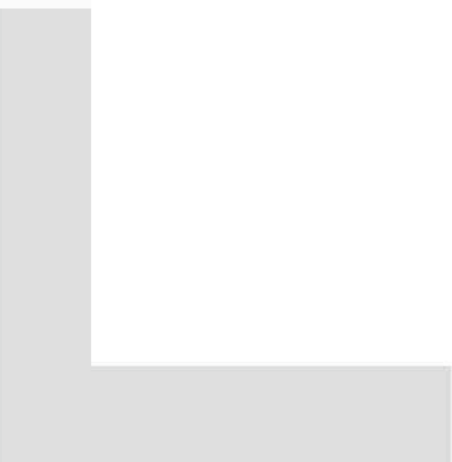
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Foreword

This is the user manual of the FRAMA-C plug-in E-ACSL¹. The content of this document corresponds to its version 0.1+dev (May 14, 2013) compatible with the version Fluorine-20130501 of FRAMA-C [2, 3]. However the development of the E-ACSL plug-in is still ongoing: features described here may still evolve in the future.

¹<http://frama-c.com/eacsl>

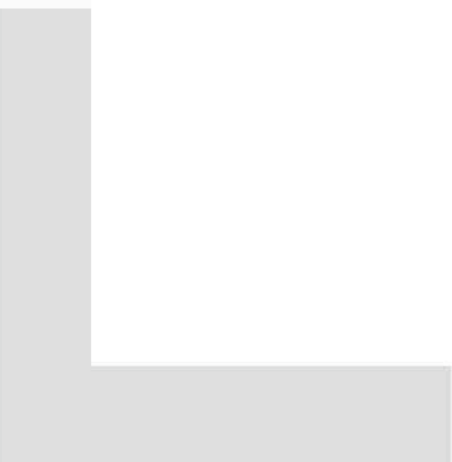


Chapter 1

Introduction

FRAMA-C [2, 3] is a modular analysis framework for the C language which supports the ACSL specification language [1]. This manual documents the E-ACSL plug-in of FRAMA-C. This plug-in automatically translates an annotated C code into a C code which fails at runtime if an annotation is violated. If no annotation is violated, the behavior of the new program is exactly the same than the one of the original program.

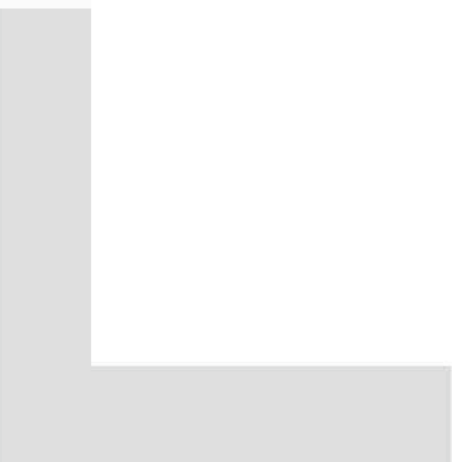
Annotations must be written in the E-ACSL specification language [5, 4] which is a subset of ACSL. This plug-in is still in a preliminary state: some parts of E-ACSL are not yet supported. Which E-ACSL annotations are currently handled by the E-ACSL plug-in is documented in a separated document [6].



Chapter 2

What the Plug-in Provides

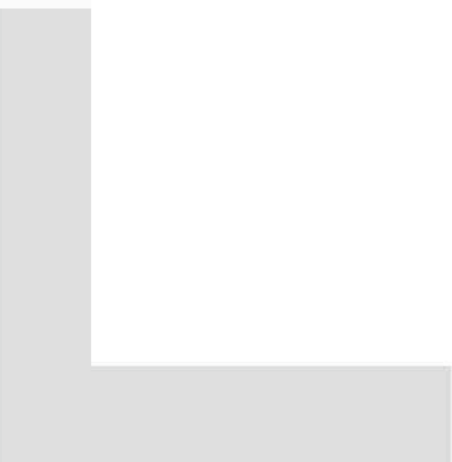
- simple example illustrating project generation and how to handle this project to effectively verify annotations at runtime with GCC. (`-e-acsl-project`)
- takes care of architecture (32, 64 bits)
- memory model (linking issue)
- GMP (linking issue)
- incomplete program
 - function without code
 - program without main
- customizing `e_acsl_assert`
- combining E-ACSL with others Frama-C analysis (`e-acsl-prepare` and `-e-acsl-valid`)
- verbosing policy



Chapter 3

Known limitations

- of course, annotations that are not yet translated [\[6\]](#)
- function without code
- recursive functions
- function pointers
- complex control flow graph



Bibliography

- [1] Patrick Baudin, Jean-Christophe Filliâtre, Claude Marché, Benjamin Monate, Yannick Moy, and Virgile Prevosto. *ACSL: ANSI/ISO C Specification Language. Version 1.7*, April 2013.
- [2] Loïc Correnson, Pascal Cuoq, Florent Kirchner, Armand Puccetti, Virgile Prevosto, Julien Signoles, and Boris Yakobowski. *Frama-C User Manual*, May April. <http://frama-c.cea.fr/download/user-manual.pdf>.
- [3] Pascal Cuoq, Florent Kirchner, Nikolai Kosmatov, Virgile Prevosto, Julien Signoles, and Boris Yakobowski. Frama-C, A software Analysis Perspective. In *Software Engineering and Formal Methods (SEFM)*, October 2012.
- [4] Mickaël Delahaye, Nikolai Kosmatov, and Julien Signoles. Common specification language for static and dynamic analysis of C programs. In *the 28th Annual ACM Symposium on Applied Computing (SAC)*, pages 1230–1235. ACM, March 2013.
- [5] Julien Signoles. *E-ACSL: Executable ANSI/ISO C Specification Language. Version 1.7*, May 2013. URL: <http://frama-c.com/download/e-acsl/e-acsl.pdf>.
- [6] Julien Signoles. *E-ACSL Version 1.7. Implementation in Frama-C Plug-in E-ACSL version 0.2*, May 2013. URL: <http://frama-c.com/download/e-acsl/e-acsl-implementation.pdf>.